

Chapter 83

Planar Tracking

This chapter provides an overview of how to use the Planar Tracker node, and how to use it to make match moves simple.

For more information about the Planar Tracker node, see *Chapter 119, “Tracking Nodes,”* in the DaVinci Resolve Reference Manual or Chapter 59 in the Fusion Reference Manual.

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Introduction to Tracking

Fusion includes three different Tracking nodes that let you analyze different kinds of motion. Once you have tracked motion on a clip, you can then use the resulting data for stabilization, motion smoothing, matching the motion of one object to that of another, and a host of other essential tasks. Each tracker type has its own chapter in this manual. This chapter covers the tracking techniques with the Planar Tracker node.

Using the Planar Tracker

The Planar Tracker node is designed to deal with match moving issues on flat, unvarying surfaces that commonly arise during post-production. Examples of flat, unvarying surfaces include clips containing a license plate, a road sign, or a brick wall that often need images merged on top of them, such as replacing the numbers in the license plate, changing the city's name in the road sign, or placing a billboard poster on the empty brick wall.

The Planar Tracker automates this process by analyzing the perspective distortions of a planar surface on a background plate over time, and then re-applying those same perspective distortions to a different foreground.

TIP: Part of using the Planar Tracker is also knowing when to give up and fall back to using Fusion's Tracker node or to manual keyframing. Some shots are simply not trackable, or the resulting track suffers from too much jitter or drift. The Planar Tracker is a time-saving node in the artist's toolbox and, while it can track most shots, no tracker is a 100% solution.

Different Ways of Using the Planar Tracker Node

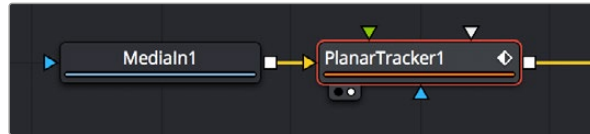
Like the other tracking nodes found in Fusion, the Planar Tracker can both analyze and contain the resulting image tracking data interior to the node, and it can also use that tracking data to transform either another image, paint strokes, or a polygon mask shape.

The Planar Tracker provides four modes of operation:

- **Track:** Used to isolate a planar surface and track its movement over time. Then, you can create a Planar Transform node that uses this data to match move another clip in various ways.
- **Steady:** After analyzing a planar surface, this mode removes all motion and distortions from the planar surface, usually in preparation for some kind of paint or roto task, prior to "unsteadying" the clip to add the motion back.
- **Corner Pin:** After analyzing a planar surface, this mode computes and applies a matching perspective distortion to a foreground image you connect to the foreground input of the Planar Tracker node, and merges it on top of the tracked footage.
- **Stabilize:** After analyzing a planar surface, allows smoothing of a clip's translation, rotation, and scale over time. Good for getting unwanted vibrations out of a clip while retaining the overall camera motion that was intended.

Setting Up to Use the Planar Tracker

Similar to the Tracker node, to do a planar track, you need to connect the output of the image you want to track to the background input of a Planar Tracker node.

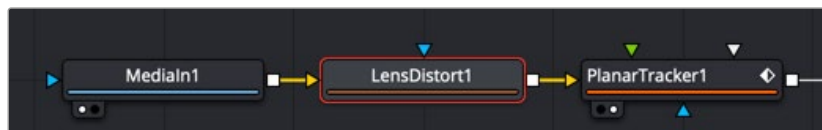


Connecting an image to the background input of a PlanarTracker node

Check for Lens Distortion

If the image has barrel distortion, or any other kinds of lens distortion, it can adversely affect your track. The more lens distortion in the footage, the more the resulting track will slide and wobble. If you can see distortion in the image or you're having problems with the track, you'll want to try inserting the Lens Distort node between the image and the Planar Tracker to eliminate this problem.

Fusion's Lens Distort node can be used to remove or add lens distortion in an image. Connecting the Medialn or Loader node to the Lens Distort node displays controls for manually correcting lens distortion. If you use Synth Eyes, PFTrack or 3D Equalizer software, you can also import lens data from those applications to make the adjustments more automatic.



A Lens Distort node inserted between a Medialn1 and Planar Tracker to remove lens distortion

For more information about using the Lens Distort node, see *Chapter 123, "Warp Nodes,"* in the DaVinci Resolve Reference Manual or Chapter 63 in the Fusion Reference Manual.

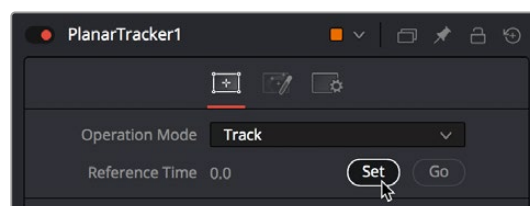
If you are using DaVinci Resolve, you can use the Lens Corrections control in the Cut page or Edit page. This adjustment carries over into the Fusion page. Lens correction in DaVinci Resolve automatically analyzes the frame in the Timeline viewer for edges that are being distorted by a wide angle lens. Clicking the Analyze button moves the Distortion slider to provide an automatic correction. From there, the Medialn node in the Fusion page will have the correction applied, and you can begin planar tracking.

A Basic Planar Tracker Match Move Workflow

Using the Planar Tracker is a process, but it's straightforward once you've learned how to use it. The following procedure tries to make this process as clear as possible.

To track a surface using the Planar Tracker:

- 1 Make sure the Operation Mode is set to Track, as you need to analyze an image to track a surface before you do anything else.
- 2 With the background input of the Planar Tracker connected to an image, and the Planar Tracker open in a viewer, move the playhead to a frame of video where the planar surface you want to track is at its largest, is unoccluded, and is clearly a plane, and then click the Set button in the Track panel of the Inspector to make this the reference frame that will be used to guide the track.



Clicking the Set button to set the reference frame to use for analysis

- 3 Next, you'll need to identify the specific pattern within the image that you want to track. In most cases, this will probably be a rectangle, but any arbitrary closed polygon can be used. The pixels enclosed by this region will serve as the pattern that will be searched for on other frames. Please note that it is important that the pattern is drawn on the reference frame. In this example, we want to track the wall behind the man, so we draw a polygon around part of the wall that the man won't pass over as he moves during the shot.



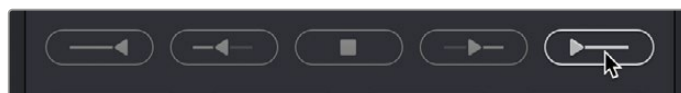
Drawing a polygon to identify the part of the image you want to track, which should be a flat, trackable plane

TIP: Do not confuse the pattern you're identifying with the region you're planning to corner pin (which always has four corners and is separately specified in Corner Pin mode).

- 4 (Optional) If moving objects partially cover up or occlude the planar surface, you may wish to connect a mask that surrounds and identifies these occlusions to the white “occlusion mask” input of the Planar Tracker. This lets the Planar Tracker ignore details that will cause problems.

When using the Hybrid Tracker, providing a mask to deal with occluding objects is nearly mandatory, while with the Point Tracker it is recommended to try tracking without a mask.

- 5 If necessary, move the playhead back to the reference frame, which in this case was the first frame. Then, click the Track To End button and wait for the track to complete.



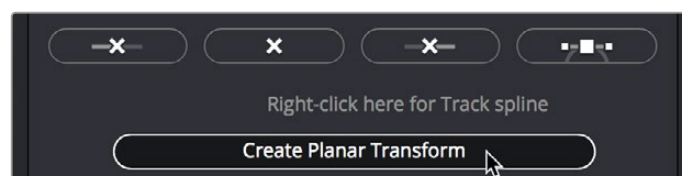
The Analyze buttons of the Planar Tracker

As the clip tracks, you can see track markers and trails (if they're enabled in the Options tab of the Inspector) that let you see how much detail is contributing to the track, and the direction of motion that's being analyzed.



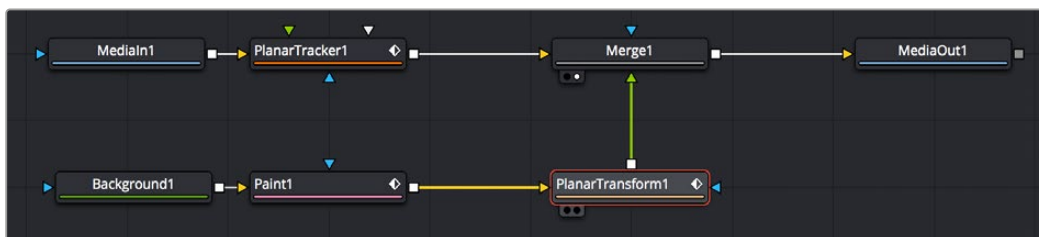
During tracking, you can see track markers and trails to follow how well the track is going

- 6 Once the track is complete, play through the clip to visually inspect the track so you can evaluate how accurate it is. Does it stick to the surface? Switching to Steady mode can help here, as scrubbing through the clip in Steady mode will help you immediately see unwanted motion in the track.
- 7 Since we're doing a match move, click the Create Planar Transform button to export a Planar Transform node that will automatically transform either images or masks to follow the analyzed motion of the plane you tracked.



Clicking Create Planar Transform to create a node to use to transform other images or masks

In this case, the Planar Transform node will be inserted after a pair of Background and Paint nodes that are being used to put some irritatingly trendy tech jargon graffiti on the wall. The Planar Transform will automatically transform the Paint node's output connected to its background input to match the movement of the wall.



Adding the PlanarTransform node after a Paint node to match move it to the background image, combining it via a Merge node

The result is a seamless match move of the fake graffiti married to the wall in the original clip.



The final result; the paint layer is match moved to the background successfully

TIP: If you want to composite semi-transparent paint strokes on the wall, or use Apply modes with paint stroke, you can attach a Paint node to a Background node set to 100 transparency. The resulting image will be whatever paint strokes you make against transparency and is easy to composite.

Tips for Choosing Good Planes to Track

The region to track is specified by drawing a polygon on the reference frame. Make sure the region selected belongs to a physically planar surface in the shot. Sometimes a region that is only approximately planar can be used. In general, the less planar the surface, the poorer the quality of the resulting track.

As a rule of thumb, the more pixels in the pattern, the better the quality of the track. In particular, this means on the reference frame, the pattern to be tracked should:

- Be as large as possible.
- Be as much in frame as possible.
- Be as unoccluded as possible by any moving foreground objects.
- Be at its maximum size (e.g., when tracking an approaching road sign, it is good to pick a later frame where it is 400 x 200 pixels rather than 80 x 40 pixels).
- Be relatively undistorted (e.g., when the camera orbits around a flat stop sign, it is better to pick a frame where the sign is face on parallel to the camera rather than a frame where it is at a highly oblique angle).

If the pattern contains too few pixels or not enough trackable features, this can cause problems with the resulting track, such as jitter, wobble, and slippage. Sometimes dropping down to a simpler motion type can help in this situation.